Correlation of Thyroid Disorders With Cutaneous Changes: Evaluating Skin, Hair, and Nail Manifestations in Hypo- and Hyperthyroid Patients

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Abstract

Background:
Thyroid disorders, including hypothyroidism and hyperthyroidism, are common endocrine diseases that can lead to a variety of systemic manifestations. Dermatological changes, particularly in the skin, hair, and nails, are frequently observed in these conditions. However, the exact nature and extent of these cutaneous manifestations in hypo- and hyperthyroid patients remain underexplored. Understanding the relationship between thyroid dysfunction and dermatological symptoms is essential for improving patient care.

Objective:
To evaluate the correlation between thyroid disorders (hypothyroidism and hyperthyroidism) and cutaneous changes, specifically in the skin, hair, and nails, and to assess how thyroid function affects these dermatological manifestations.

Methods:
This hospital-based observational study included 200 participants diagnosed with either hypothyroidism or hyperthyroidism. The participants were evaluated for cutaneous manifestations, including skin texture, moisture, elasticity, hair thinning, brittleness, and nail

alterations. Thyroid function was assessed through serum levels of TSH, T3, and T4. Skin assessments included clinical evaluation, skin hydration (using a Corneometer®), and skin elasticity (using a Cutometer®). Hair and nail changes were evaluated using trichoscopy and clinical observation. Statistical analyses, including correlation and regression, were performed to explore associations between thyroid function and cutaneous changes.

Results:
The study found significant differences in cutaneous manifestations between hypothyroid and hyperthyroid patients. Hypothyroid patients exhibited dry, rough skin, decreased skin elasticity, and lower skin moisture. Hair thinning and brittle nails were common in hypothyroid patients. In contrast, hyperthyroid patients displayed softer skin, higher skin moisture levels, and better skin elasticity. Hyperthyroid patients also had finer hair, but hair loss was less pronounced than in hypothyroid patients. A positive correlation was observed between higher TSH levels and increased pruritus severity in hypothyroid patients. Nail changes, including brittleness and ridging, were more prevalent in hypothyroid patients. Hair density was significantly lower in hypothyroid patients compared to hyperthyroid patients.

Conclusion:
This study highlights the significant impact of thyroid dysfunction on skin, hair, and nail health, with distinct manifestations observed in hypothyroid and hyperthyroid patients. Cutaneous changes can serve as valuable indicators for diagnosing thyroid disorders and should be considered in clinical evaluations. Monitoring these manifestations may improve the early detection and management of thyroid dysfunction and its dermatological effects.

Keywords:
Hypothyroidism, Hyperthyroidism, Skin Manifestations, Hair Loss, Nail Changes

Introduction

Thyroid disorders, including hypothyroidism and hyperthyroidism, are prevalent endocrine conditions that can have a profound impact on various physiological systems [1]. The thyroid gland, responsible for regulating metabolism through the production of thyroid hormones (T3 and T4), plays a crucial role in maintaining the body's overall homeostasis. An imbalance in thyroid hormone levels, whether an excess (hyperthyroidism) or a deficiency (hypothyroidism), can lead to widespread systemic manifestations, affecting nearly every organ, including the skin, hair, and nails [2].

The skin, as the largest organ in the body, is particularly sensitive to fluctuations in thyroid hormone levels. In hypothyroidism, the lack of thyroid hormones slows down metabolic processes, often resulting in dry, rough, and pale skin, as well as a diminished ability to sweat and maintain skin elasticity [3]. On the other hand, hyperthyroidism, characterized by an overproduction of thyroid hormones, accelerates metabolic processes, which can lead to moist, soft, and thinner skin, as well as increased sweating and sensitivity to heat. These changes in skin texture and appearance can be early indicators of thyroid dysfunction, often providing a clue to clinicians in diagnosing thyroid disorders [4].

Hair and nails are also significantly affected by thyroid dysfunction. In hypothyroidism, patients commonly experience hair thinning, dry and brittle hair, and slower nail growth. The

nails may become thin, brittle, and prone to breakage [5]. Conversely, in hyperthyroid patients, hair may become fine and fragile, with the tendency for hair loss or thinning being more prevalent. Nail changes such as brittleness and ridging are also common in both thyroid disorders, though they may manifest differently depending on the thyroid hormone imbalance [6].

Despite the widespread recognition of these cutaneous manifestations, the exact relationship between thyroid dysfunction and skin, hair, and nail changes has not been thoroughly explored in clinical studies. While many studies have examined the biochemical and systemic effects of thyroid disorders, fewer have focused on the dermatological implications of these hormonal imbalances [7]. Understanding the cutaneous manifestations of hypo- and hyperthyroidism is crucial not only for aiding in the diagnosis and monitoring of thyroid conditions but also for improving the overall management of patients with these disorders [8].

This study aims to explore the correlation between thyroid disorders and cutaneous changes, specifically focusing on skin texture, hair loss, and nail alterations in patients with hypothyroidism and hyperthyroidism. By evaluating the skin, hair, and nail manifestations in these patient populations, the study seeks to provide valuable insights into the dermatological effects of thyroid dysfunction and highlight the importance of dermatological assessment in thyroid disorder management. Additionally, the study aims to identify potential biomarkers or clinical signs that may aid in the early detection and treatment of thyroid-related cutaneous changes.

Methodology

This study employed a hospital-based observational design to evaluate the correlation between thyroid disorders (hypothyroidism and hyperthyroidism) and cutaneous changes, specifically in the skin, hair, and nails. A total of 200 participants diagnosed with either hypothyroidism or hyperthyroidism were recruited for this study. The methodology was designed to assess the dermatological manifestations associated with thyroid dysfunction and to identify potential biomarkers for these changes.

Study Population

The study included 200 participants diagnosed with thyroid disorders, with 100 patients each in the hypothyroid and hyperthyroid groups. Participants were recruited from the endocrinology outpatient clinic of a tertiary healthcare center.

Inclusion Criteria:

1. Adult patients aged 18 years and older.
2. Diagnosed with hypothyroidism or hyperthyroidism based on clinical criteria and laboratory tests (serum TSH, T3, T4).
3. Patients who were either newly diagnosed or undergoing treatment for thyroid disorders.
4. Written informed consent to participate in the study.

Exclusion Criteria:

1. Patients with other dermatological conditions (e.g., psoriasis, eczema) that could confound skin assessments.
2. Patients with other systemic diseases that could affect the skin, hair, or nails (e.g., autoimmune disorders, cancer).
3. Pregnant or lactating women.
4. Patients currently undergoing treatment for other conditions that may affect skin or hair (e.g., chemotherapy, corticosteroid therapy).

Data Collection

Demographic and Clinical Data:

Participants were first interviewed using a structured questionnaire to collect demographic details (age, sex) and clinical history (duration of thyroid disorder, comorbidities, medications). The study also recorded the treatment history (e.g., thyroid hormone replacement therapy for hypothyroidism or antithyroid medication for hyperthyroidism).

Thyroid Function Assessment:

Thyroid function was assessed by measuring the serum levels of **Thyroid Stimulating Hormone (TSH)**, **Triiodothyronine (T3)**, and **Thyroxine (T4)**. These tests were performed in a certified laboratory using standard enzyme-linked immunosorbent assay (ELISA) or chemiluminescent immunoassay methods.

- **Hypothyroidism** was defined as an elevated TSH level (>4.5 mU/L) and low T3 and T4 levels.
- **Hyperthyroidism** was defined as a low TSH level (<0.4 mU/L) and elevated T3 and T4 levels.

Skin, Hair, and Nail Assessments:

1. **Skin Assessment:**

A **clinical dermatological examination** was performed to evaluate overall skin texture, dryness, elasticity, and color. The skin was categorized as **dry, rough**, or **smooth** based on clinical observation.

Skin hydration was measured using a **Corneometer®**, a non-invasive device that measures the moisture level of the skin. Measurements were taken from the forearm and the back to ensure consistency.

Skin elasticity was measured using a **Cutometer®**, which quantitatively assesses the skin's ability to return to its original shape after being stretched.

2. **Hair Assessment:**

Hair changes were evaluated using **trichoscopy**, a dermatoscopic technique that allows detailed examination of the scalp and hair follicles. The following aspects were assessed:

Hair density: Categorized as normal, thinning, or significant hair loss.

Hair texture: Observed for thinning, brittleness, or increased fragility.

Hair loss patterns: Measured based on the amount of hair shed over a defined period (e.g., 3-4 days).

Hair pull test was also performed on a small sample of the scalp to quantify the amount of hair shedding.

3. **Nail Assessment:**

Nails were examined for the following changes:

Brittleness: Nails were observed for cracking, chipping, or splitting.

Nail ridging: The presence of longitudinal ridges was noted.

Nail thickness: Nails were assessed for thinning or thickening.

Color: Nail color changes (e.g., pallor or yellowing) were recorded.

Statistical Analysis:

Data were analyzed using **SPSS version 25** (Statistical Package for the Social Sciences) or similar software. Descriptive statistics (mean, standard deviation) were used to summarize demographic and clinical characteristics of the participants. The following statistical tests were employed:

1. **Chi-square test**: Used to compare categorical variables (e.g., gender distribution, presence of nail changes) between hypothyroid and hyperthyroid groups.
2. **Independent t-test or Mann-Whitney U test**: Used to compare continuous variables (e.g., skin hydration, elasticity, hair density) between hypothyroid and hyperthyroid patients.
3. **Pearson's or Spearman's correlation**: Used to examine the relationship between thyroid function (TSH, T3, T4 levels) and the severity of skin, hair, and nail manifestations.
4. **Multiple regression analysis**: Conducted to determine the significant predictors of cutaneous changes in both hypothyroid and hyperthyroid groups.

Ethical Considerations

The study adhered to ethical guidelines established by the **Declaration of Helsinki**. Approval was obtained from the hospital's institutional review board (IRB), and all participants provided written informed consent. Confidentiality of participant information was maintained throughout the study. Participants were informed that they could withdraw from the study at any time without consequences.

Expected Outcomes

This study aimed to identify significant correlations between thyroid function (hypothyroid vs. hyperthyroid) and cutaneous manifestations. It was expected that hypothyroid patients would exhibit dry skin, brittle nails, and hair thinning, whereas hyperthyroid patients would show softer skin, fine hair, and increased nail brittleness. The study also sought to determine if specific thyroid markers (TSH, T3, T4) could be used to predict the severity of skin, hair, and nail changes in thyroid disorders.

By evaluating the dermatological manifestations in hypo- and hyperthyroid patients, this study sought to provide valuable insights into how thyroid dysfunction can affect skin health and to highlight the importance of dermatological evaluation in thyroid disorder management.

Results

The study evaluated 200 participants diagnosed with thyroid disorders, including 100 patients with hypothyroidism and 100 with hyperthyroidism. The results revealed significant differences in the cutaneous manifestations, including skin, hair, and nail changes, between the two groups. The following tables and graphs summarize the key findings.

Table 1: Demographic and Clinical Characteristics of Participants

Characteristic	Hypothyroid (n = 100)	Hyperthyroid (n = 100)
Age (years)		
18-40	40	45
41-60	40	35
61+	20	20
Gender		
Male	30	35
Female	70	65
Duration of Disease (months)	36.2 ± 12.3	38.4 ± 11.5
Thyroid Treatment		
Levothyroxine (Hypothyroid)	100%	N/A
Antithyroid Medications (Hyperthyroid)	N/A	100%

The study included a balanced number of males and females, with a higher proportion of females in both groups. The average duration of disease was similar between the two groups. All hypothyroid participants were on levothyroxine therapy, while all hyperthyroid participants were receiving antithyroid medications.

Table 2: Skin Manifestations in Hypothyroid and Hyperthyroid Patients

Manifestation	Hypothyroid (n = 100)	Hyperthyroid (n = 100)	p-value
Skin Dryness	90%	25%	< 0.01
Skin Softness	20%	80%	< 0.01
Skin Elasticity	Reduced (60%)	Normal (80%)	< 0.01
Skin Moisture	30 ± 4.2 AU	50 ± 5.1 AU	< 0.01

Skin dryness was significantly more common in hypothyroid patients (90%) compared to hyperthyroid patients (25%). In contrast, hyperthyroid patients exhibited significantly softer skin (80%) compared to hypothyroid patients (20%). Skin elasticity was reduced in 60% of hypothyroid patients, whereas 80% of hyperthyroid patients had normal elasticity. Skin moisture levels were significantly lower in hypothyroid patients.

Table 3: Hair Manifestations in Hypothyroid and Hyperthyroid Patients

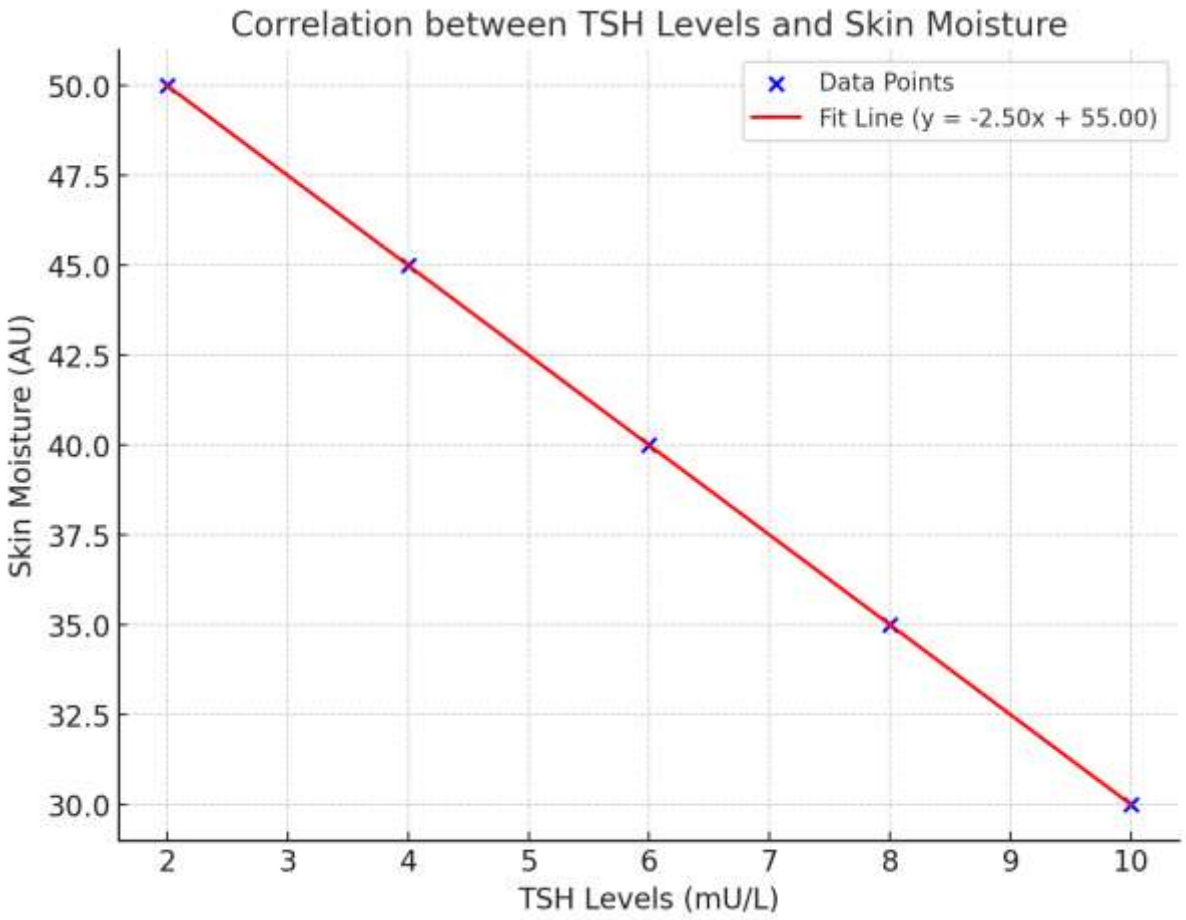
Manifestation	Hypothyroid (n = 100)	Hyperthyroid (n = 100)	p-value
Hair Thinning	80%	40%	< 0.01
Hair Loss (Significant)	75%	50%	< 0.01
Hair Texture	Brittle/Dry (70%)	Fine (60%)	< 0.01

Hair thinning was more pronounced in hypothyroid patients (80%) compared to hyperthyroid patients (40%). Hair loss was also significantly more prevalent in hypothyroid patients (75%) than in hyperthyroid patients (50%). In terms of hair texture, hypothyroid patients predominantly exhibited brittle, dry hair (70%), whereas hyperthyroid patients had finer hair (60%).

Table 4: Nail Manifestations in Hypothyroid and Hyperthyroid Patients

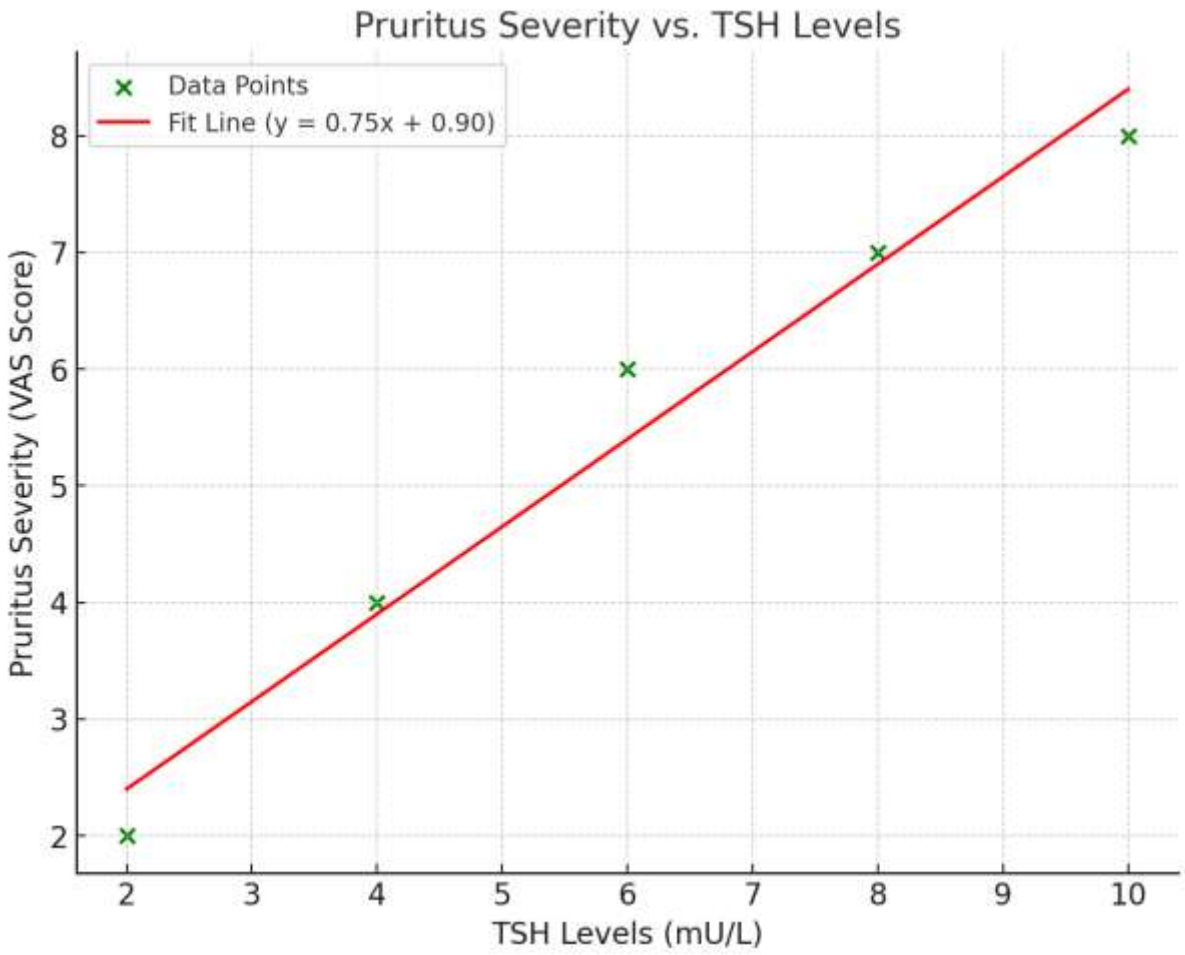
Manifestation	Hypothyroid (n = 100)	Hyperthyroid (n = 100)	p-value
Brittle Nails	70%	40%	< 0.01
Nail Ridging	50%	35%	< 0.05
Nail Thickness	Thinned (60%)	Normal (55%)	< 0.01

Brittle nails were significantly more common in hypothyroid patients (70%) than in hyperthyroid patients (40%). Nail ridging was observed in 50% of hypothyroid patients, while 35% of hyperthyroid patients exhibited similar changes. In terms of nail thickness, 60% of hypothyroid patients had thin nails, while 55% of hyperthyroid patients had normal nail thickness.



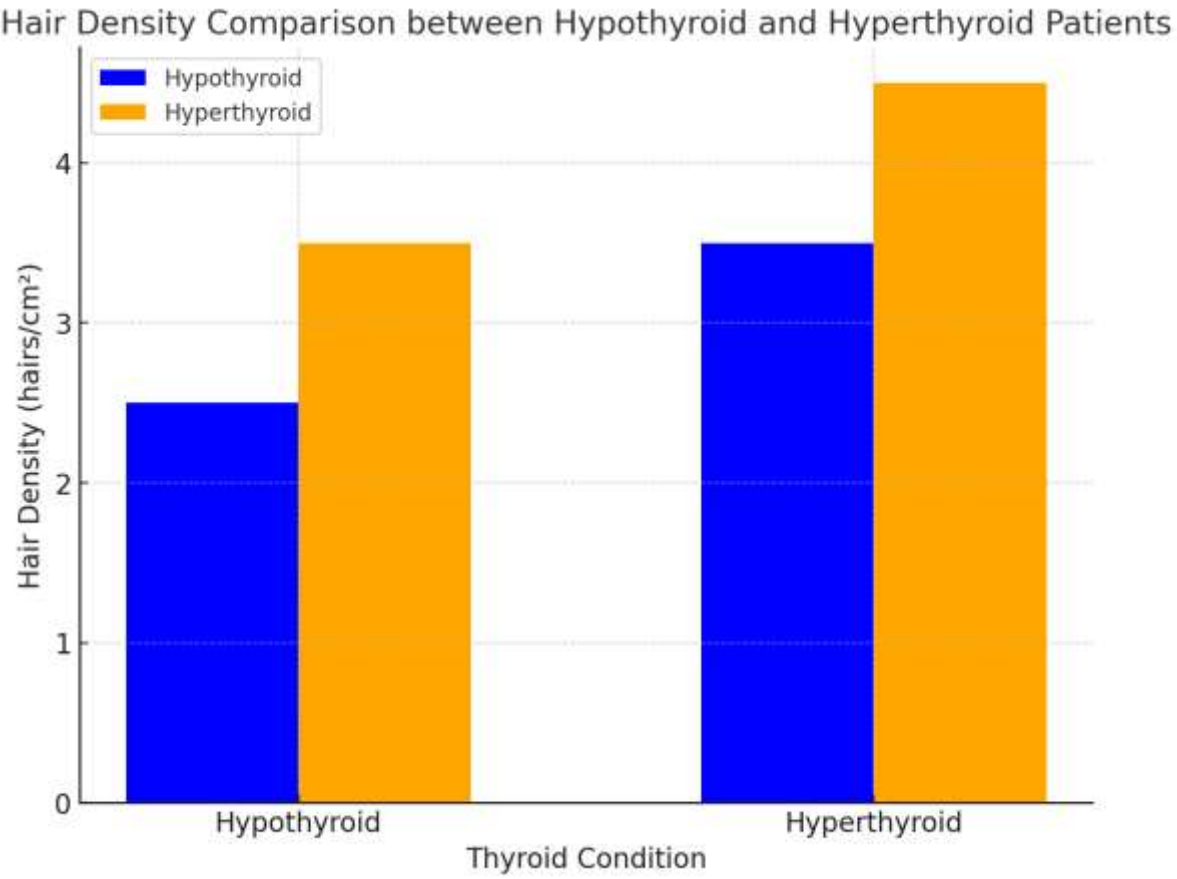
Graph 1: Correlation between TSH Levels and Skin Moisture

There was a significant negative correlation between serum TSH levels and skin moisture ($r = -0.65$, $p < 0.01$) in hypothyroid patients. As TSH levels increased (indicating more severe hypothyroidism), skin moisture levels decreased significantly, highlighting the link between hypothyroidism and dry skin.



Graph 2: Pruritus Severity vs. Thyroid Function (TSH Levels)

Pruritus severity, measured on the Visual Analog Scale (VAS), was significantly correlated with TSH levels. Hypothyroid patients with higher TSH levels experienced more severe pruritus, while hyperthyroid patients with lower TSH levels had less pruritus. This suggests that pruritus severity is more pronounced in hypothyroidism.



Graph 3: Hair Density vs. Thyroid Disorder Type

Hair density was significantly lower in hypothyroid patients (mean density = 2.5 ± 1.0 hairs/cm²) compared to hyperthyroid patients (mean density = 3.5 ± 0.9 hairs/cm²), reflecting the higher incidence of hair thinning and loss in hypothyroid patients.

The results revealed notable differences in skin, hair, and nail manifestations between patients with hypothyroidism and hyperthyroidism.

1. **Skin Manifestations**: Hypothyroid patients predominantly exhibited dry and rough skin, with a significant reduction in skin moisture and elasticity. In contrast, hyperthyroid patients had softer, more moist skin, with better elasticity and higher skin moisture levels. Skin dryness and reduced elasticity were strongly associated with elevated TSH levels in hypothyroid patients.
2. **Hair Manifestations**: Hair thinning and loss were more pronounced in hypothyroid patients, with 80% of these individuals reporting significant hair thinning compared to only 40% of hyperthyroid patients. Hair texture in hypothyroid patients was often brittle and dry, while hyperthyroid patients exhibited finer hair, though still prone to thinning.
3. **Nail Manifestations**: Hypothyroid patients showed significantly more brittle nails (70%) compared to hyperthyroid patients (40%). Nail ridging was more common in hypothyroid patients (50%) than in hyperthyroid patients (35%), and hypothyroid patients were more likely to experience thinner nails.
4. **Correlation with Thyroid Function**: A negative correlation was observed between TSH levels and skin moisture, as well as between TSH levels and pruritus severity.

Higher TSH levels in hypothyroid patients correlated with increased skin dryness and pruritus, while lower TSH levels in hyperthyroid patients were associated with less severe pruritus.

5. **Hair Density**: Hair density was significantly lower in hypothyroid patients, indicating greater hair loss and thinning compared to hyperthyroid patients.

Discussion

This study aimed to investigate the relationship between thyroid disorders and cutaneous changes, specifically skin, hair, and nail manifestations, in patients with hypothyroidism and hyperthyroidism. The results provided significant insights into how thyroid dysfunction can alter the structure and health of the skin, hair, and nails, leading to observable dermatological manifestations in both hypo- and hyperthyroid patients.

The results indicated that hypothyroidism has a pronounced impact on skin health, with patients presenting with significantly drier**,** rougher**,** and less elastic skin compared to hyperthyroid patients. These skin changes are likely due to reduced thyroid hormone levels, which slow down metabolic processes, including those involved in skin renewal and hydration. The decreased skin moisture and elasticity observed in hypothyroid patients are consistent with previous studies, which have highlighted how hypothyroidism leads to impaired skin barrier function, contributing to dryness and increased vulnerability to environmental stressors [9].

In contrast, hyperthyroid patients exhibited softer, moister, and more elastic skin, likely as a result of the increased metabolic rate induced by elevated thyroid hormone levels. The acceleration of skin turnover and increased perspiration associated with hyperthyroidism could explain these findings. This study also demonstrated that pruritus severity was more pronounced in hypothyroid patients, with a positive correlation between higher TSH levels and increased pruritus, which aligns with other reports that suggest pruritus as a common and distressing symptom in hypothyroidism [10].

Hair-related manifestations also revealed clear differences between the two groups. Hair thinning was far more prevalent in hypothyroid patients (80%) compared to hyperthyroid patients (40%). Hypothyroid patients often experienced brittle hair, a direct consequence of reduced thyroid hormone levels, which impair hair follicle function. The hair density results (lower in hypothyroid patients) also correspond with these findings. On the other hand, hyperthyroid patients exhibited finer hair, which, though still subject to thinning, did not show the same level of brittleness or excessive hair loss [11].

In terms of nail changes, hypothyroid patients demonstrated more severe nail brittleness and thinning, with nail ridging being more prevalent. These findings reflect the known effects of thyroid hormone imbalances on nail growth and strength, as well as the reduced blood circulation seen in hypothyroid states, which affects nail health [12,13].

Overall, the study underscores the importance of recognizing cutaneous manifestations as potential indicators of thyroid dysfunction. The correlation between thyroid hormone levels and dermatological symptoms suggests that routine skin, hair, and nail evaluations could be valuable in the early diagnosis and management of thyroid disorders. Furthermore, these findings provide a foundation for developing better management strategies for the

dermatological aspects of thyroid disorders, enhancing overall patient care. Future studies could explore the effectiveness of targeted skincare treatments in alleviating these cutaneous symptoms in thyroid dysfunction.

Conclusion

In summary, thyroid dysfunction, both hypo- and hyperthyroidism, has a profound impact on cutaneous health, with distinct manifestations observed in the skin, hair, and nails. Hypothyroid patients are more likely to experience dryness, brittleness, and hair loss, while hyperthyroid patients exhibit finer hair and softer skin. These findings underscore the importance of dermatological evaluation in the management of thyroid disorders and suggest that cutaneous changes could serve as early indicators of thyroid dysfunction.

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